

Logic and Proofs

Trying a new typesetting format for this topic, lmk if you end up preferring this format or the previous format (or dont care about it)

Logic

- A statement in math can be either true or false (has a truth value of T or F). We often call it by a capital variable as shorthand
 - P : "The sky is blue right now"
 - Q : "It is night time"
 - These are two statements, each is either true or false, often called *primitive statements*
- We can construct more complicated statements out of these primitives, using *connectives*. They connect primitive statements together
 - The statement S : "it is day time" should have the exact opposite truth value to Q
 - when its night: $Q = T$ and $S = F$
 - when its day: $Q = F$ and $S = T$
- when we create a new statement out of primitive statements and connectives, we call them *compound statements*

Connectives

1. **Negation**: negation of a statement p is the statement 'not p ' and is denoted by $\sim p$, the truth value of $\sim p$ is opposite of p
2. **"and"** (conjunction) : basically "and" conjunction of p and q is the statement $p \wedge q$ which is true if only *both* p and q are true
3. **"or"** (disjunction): basically "or", disjunction of p and q is the statement $p \vee q$ which is true if either *or* both p and q are true
 - Note: "Or" here works slightly different to English language. This is the inclusive or, meaning we accept the possibility of both statements being true
4. **implication**: The statement that p implies q , which defaults to true *except for the case* $T \implies F$
This means that the statement $p \implies q$ is true even if $p = F$, regardless of the value of q as it does not violate the statement
 - This means that the statement $p \implies q$ being true does not imply anything about the values of p or q
 - **Converse** of an implication $p \implies q$ is the statement $q \implies p$, i.e just swap the primitive statements around

5. Double implication: $p \iff q$, which is equivalent to $(p \implies q) \wedge (q \implies p)$

- Examples:
 - P : I am wearing a jacket
 - Q : It is winter
- 1. $\sim P$: I am not wearing my jacket
- 2. $P \wedge Q$: I am wearing my jacket and it is winter
- 3. $P \vee Q$: I am wearing my jacket or it is winter
 - This is NOT the same as "**either** I am wearing my jacket or it is winter". Why?
 - $P \vee Q$ is true if any of P or Q is true, even if both of them are true!
- 4. $Q \implies P$: If it's winter, I am wearing my jacket
 - it's reasonable to say that $Q \implies P$ is a true statement. Whenever it's cold, I'll wear my jacket. But is $P \implies Q$ true?
 - The converse of a true implication is not always true...
- Remember, there is no rule saying I must use variables to represent my statements. Sometimes it's clearer not to
 - Let $A = \{1, 2\}$, $B = \{2, 3\}$
 - Translate this into formal logic: "2 is in A and B "

Quantifiers

- Way to quantify "how many"
 - $\forall$: for all (every one of)
 - $\exists$: there exists (at least one of)
- Often, "such that" is abbreviated for "s.t" after using the quantifiers
- Examples: say $S = \{2, 4, 5, 6, 10\}$. We could say the following things about this set
 - $\forall x \in S, x > 1$
 - "For all elements x in S , x is bigger than one"
 - $\exists y \in S$ s.t y is odd
 - "There exists a value y in S such that y is odd"
- Examples: describe what we would have to demonstrate in order to disprove each of the following statements
 - At every school in Melbourne, there is a student who is at least seven feet tall
 - For all schools in Melbourne, there exists a teacher who gives every student a grade of either A or B
 - There exists a school in Melbourne where every student is at least six feet tall